FUEL ANALYSIS REPORT

REPORT ID: S260924360	**CUSTOMER:** HUAYU SHIP MANAGEMENT LIMITED	**SHIP(IMO NO):** XINSHUN(9252199)	**BUNKER PORT:** MALTA

CONFORMANCE

The fuel sample tested conforms to Table 1 of ISO 8217:2005 specifications for grade MGO - DMA LS

*Conformance to table 1 or table 2 of ISO 8217 does not imply conformance to other sections of ISO 8217 such as para 5.

BUNKER INFORMATION		BDN/CLIENT INFORMATION		SAMPLE DETAILS	
BUNKER DATE	25-Sep-2026	**GRADE**	DMA LS	**SENT FROM**	
BUNKER PORT	MALTA	**DENSITY@15°C kg/m³**	883.4	**SENT DATE**	29-Sep-2026
SUPPLIER	ORIM ENERGY	**VISCOSITY@40°C cSt**	5.937	**REPORT DATE**	12-Oct-2026
BARGE	MT ALIKI	**SULFUR %m/m**	0.095	**TEST DATE**	11-Oct-2026
SAMPLING POINT	MANIFOLD	**WATER %v/v**	-	**RETENTION DATE**	11-ONov-2026
SAMPLING METHOD	DRIP	**QUANTITY (MT)**	195.385	**PO NUMBER**	-

SEAL INFORMATION

LAB SEAL	MARPOL SEAL	SHIP SEAL	BARGE SEAL
D593354(Sealed)	D593351		

RECOMMENDED TEMPERATURES

A minimum viscosity has to be maintained at engine inlet and for practical purpose some engine makers recommend this to be not less than 2cSt. Based on tested viscosity, this oil should not be heated above 78°C temperature in order to meet the requirement.

CALCULATIONS OF QUANTITY SUPPLIED BASED ON LAB TESTS

	UNITS	BUNKER INFORMATION	TESTED RESULTS
Density @15°C	**kg/m³**	883.4	882.9
Water	**%v/v**	-	0.000
Quantity supplied as per ship	**MT**	195.385	
Quantity shortage due to density	**MT**	0.110	
Quantity shortage due to water	**MT**	-	
Total Quantity Shortage	**MT**	0.110	

*Above figures do not account for ISO 4259 criteria given in ISO 8217 Annex F. (This is not applicable for delivery made by mass flowmeter)

TEST DESCRIPTION	UNITS	RESULTS	ISO 8217 LIMITS	METHOD
Density @15°C	kg/m³	882.9	Max 890.0	ISO 12185
Viscosity @ 40°C	cSt	5.920	Min 1.5 Max 6.0	ISO 3104
Flash Point	°C	> 70.0	Min 60.0	ISO 3679
Pour Point	°C	0	Max(Summer) 0 Max(Winter) -6	ISO 3016
Micro Carbon Residue	% m/m	0.01	Max 0.30	ISO 10370
Ash	% m/m	0.003	Max 0.010	Calculated
Water	% v/v	< 0.01	-	ASTM D6304
Sulfur	% m/m	0.093	Max 0.10	ISO 8754
Vanadium	mg/kg	< 1	-	VLS-LM8
Al+Si (Catfines)	mg/kg	2	-	VLS-LM8
Sodium	mg/kg	< 1	-	VLS-LM8
Aluminum	mg/kg	< 1	-	VLS-LM8
Silicon	mg/kg	< 1	-	VLS-LM8
Iron	mg/kg	< 1	-	VLS-LM8
Nickel	mg/kg	< 1	-	VLS-LM8
Calcium+	mg/kg	< 1	-	VLS-LM8
Phosphorous+	mg/kg	< 1	-	VLS-LM8
Zinc+	mg/kg	< 1	-	VLS-LM8
Potassium	mg/kg	< 1	-	VLS-LM8
Lead	mg/kg	< 1	-	VLS-LM8
Magnesium	mg/kg	< 1	-	VLS-LM8
API Gravity	-	29.05	-	-
Net Specific Energy	MJ/kg	41.890	-	Calculated
Gross Specific Energy	MJ/kg	45.350	-	Calculated
Visual Appearance		Clear & Bright	-	LM**

LM** - LAB METHOD

Calcium > 30, Zinc > 15 and Phosphorus > 15 (ISO 8217:2005 standard)

VESSEL PAST BUNKERING HISTORY DATA					
		S260924360			
BUNKER DATE		25 – Sep – 2026			
BUNKER PORT		MALTA			
QUANTITY (MT)		195.385			
TEST DESCRIPTION	UNITS				
Density @15°C	kg/m³	882.9			
Viscosity @ 40°C	cSt	5.920			
Flash Point	°C	> 70.0			
Pour Point	°C	0			
Micro Carbon Residue	% m/m	0.01			
Ash	% m/m	0.003			
Water	% v/v	< 0.01			
Sulfur	% m/m	0.093			
Vanadium	mg/kg	< 1			
Al+Si (Catfines)	mg/kg	2			
Sodium	mg/kg	< 1			
Aluminum	mg/kg	< 1			
Silicon	mg/kg	< 1			
Iron	mg/kg	< 1			
Nickel	mg/kg	< 1			
Calcium	mg/kg	< 1			
Phosphorous	mg/kg	< 1			
Zinc	mg/kg	< 1			
Potassium	mg/kg	< 1			
Lead	mg/kg	< 1			
Magnesium	mg/kg	< 1			
API Gravity		29.05			
Net Specific Energy	MJ/kg	41.890			
Gross Specific Energy	MJ/kg	45.350			
Visual Appearance		Clear & Bright			

On behalf of Viswa Lab

Signed:

S.Vkl

VIGNESH KUMAR S

Date: 12-Oct-2026

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